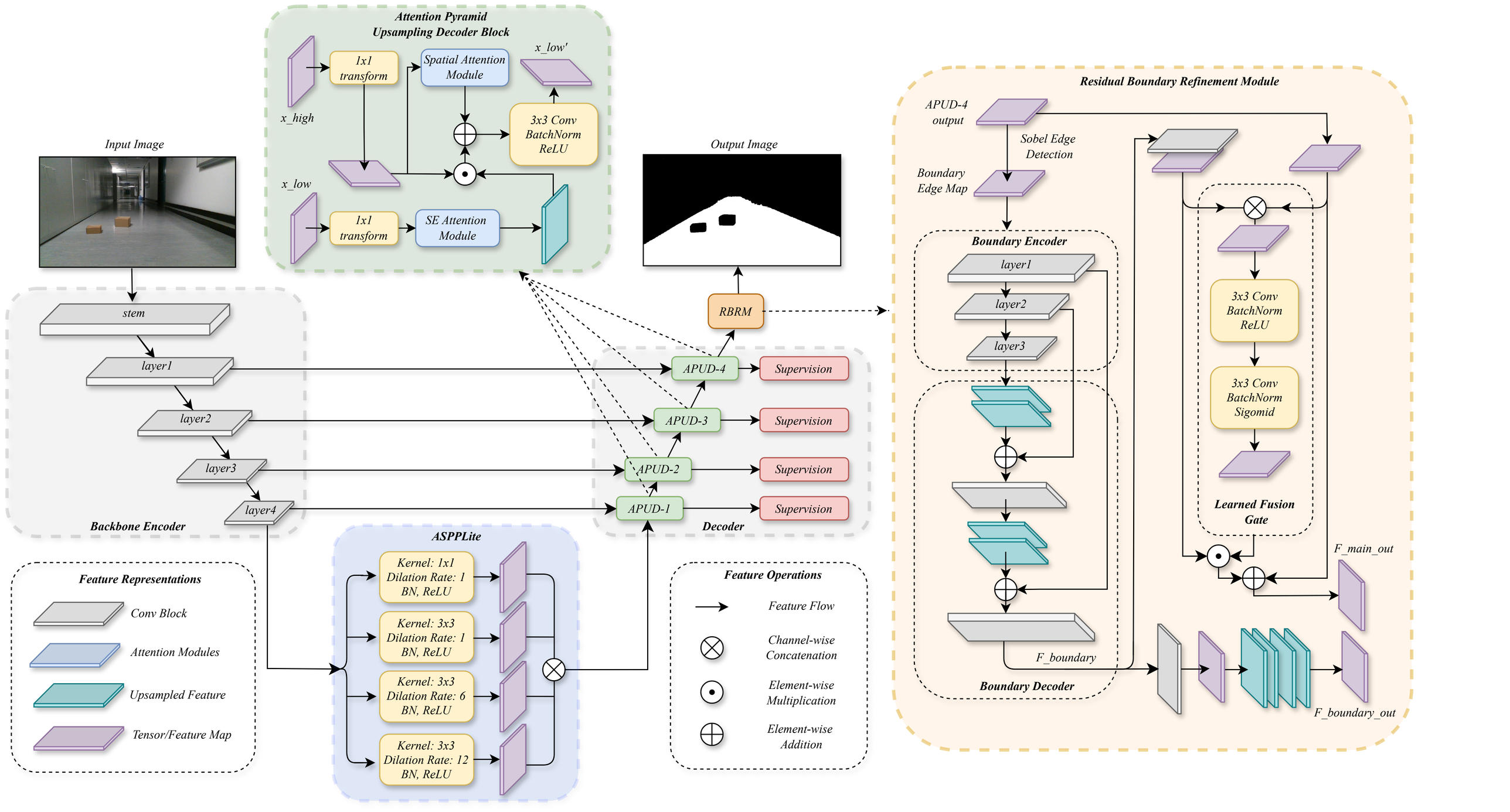
Attention Pyramid Upsampling Decoder Block
x_high
1x1 transform
Spatial Attention Module
x_low'
3x3 Conv BatchNorm ReLU
x_low
1x1 transform
SE Attention Module
Input Image
Output Image
stem
layer1
layer2
layer3
layer4
Backbone Encoder
RBRM
APUD-4
APUD-3
APUD-2
APUD-1
Supervision
Supervision
Supervision
Supervision
Decoder
ASPPLite
Kernel: 1x1 Dilation Rate: 1 BN, ReLU
Kernel: 3x3 Dilation Rate: 1 BN, ReLU
Kernel: 3x3 Dilation Rate: 6 BN, ReLU
Kernel: 3x3 Dilation Rate: 12 BN, ReLU
Feature Representations
Conv Block
Attention Modules
Upsampled Feature
Tensor/Feature Map
Feature Operations
Feature Flow
Channel-wise Concatenation
Element-wise Multiplication
Element-wise Addition
Residual Boundary Refinement Module
APUD-4 output
Sobel Edge Detection
Boundary Edge Map
Boundary Encoder
layer1
layer2
layer3
F_boundary
Boundary Decoder
3x3 Conv BatchNorm ReLU
3x3 Conv BatchNorm Sigomid
Learned Fusion Gate
F_main_out
F_boundary_out